Small fish

While carriers of narcotics are caught in the net, big sharks operate freely

Operation Trident, a multi-agency crackdown on an international syndicate operating across the India-Myanmar-Bangladesh corridor, marks a significant step-up in the government's campaign against drug trafficking. The Union Home Minister, Amit Shah, has set December 31, 2029, as the deadline to eradicate drug networks across the country. The October 23 raids, which led to 18 people being arrested, were based on NCB investigations following the arrest of an alleged Myanmar-based kingpin in Delhi this May. However, the dismantling of the entire network would require identification of all suppliers in Myanmar and their successful prosecution. The government's strategy, involving States and Union Territories, over 40 Ministries, central agencies and the pharmaceutical sector, is outlined in the Vision Document on Drug Control (2026-2029), launched by Mr. Shah at the Narco-Coordination Centre meeting on June 26, 2026. It envisages dismantling 100 major cartels and improving prosecution outcomes. The broad strategy reflects the complexity of the problem and signals a more focused approach. Data for 2025 also show an increase of over 50% in arrests and a nearly 53% rise in cases registered under the Narcotic Drugs and Psychotropic Substances (NDPS) Act compared with 2024.

However, the government's action-plan will depend on consistent implementation and efficient coordination among agencies, and striking at the real brains behind it all. In several cases, foreign-based key operatives still remain at large and big sharks often do not fall in the net. The 2021 Mundra port case in Gujarat, involving the seizure of about 3,000 kg of heroin, is a case in point wherein the principal Afghanistan-based suppliers have yet to be brought to book. Also, the quantity of drugs seized does not, by itself, indicate whether drug availability or consumption has declined. Punjab remains a prominent example of the continuing public health challenges associated with drug abuse. The alleged drug-related killings of anti-drug activists in Mansa and Gurdaspur, along with the alleged death by suicide of a father in Sangrur who had raised concerns about the easy availability of synthetic drugs in his locality, underscore the human cost of the problem. These incidents also raise questions about the protection of those who speak out against drug trafficking and networks. These concerns lend urgency to the objective of dismantling entire networks and apprehending the "big sharks", rather than concentrating disproportionately on individual carriers. The key is to identify the ultimate beneficiaries by meticulously following the money trail and ensuring their conviction.

Fouled waters

River pollution from industrial effluents indicates state failure

The Supreme Court has pulled up, once again, the Rajasthan government over industrial effluents flowing into the Jojari river and the deadly risk it poses for the people depending on it. The problem is decades old – public interest litigation in the Rajasthan High Court alerted State machinery to the problem in 2011 – indicating an oversight by the State bordering on the systematic. The Jojari, the Luni, and the Bandi are seasonal rivers in the State's arid western region, flowing through Jodhpur, Balotra, and Pali districts. This region has also evolved as a textile-dyeing and steelworking hub, and untreated or insufficiently treated effluents from these factories have run directly into the rivers and soil, endangering the health of around 20 lakh local residents. The region also suffers rampant illegal sand mining in riverbeds. The sewage treatment plants have been able to accommodate only around half of the sewage generated in the three urban centres, according to the State's submissions to the Court in 2025, even as the Court recorded substantial underutilisation at common effluent treatment plants. The combination is suggestive of industrial sources bypassing treatment facilities, poor connections between the sources and the facilities, and/or insufficient capacity at particular sites or in specific periods.

The Court initiated a *suo motu* writ petition in 2025 and constituted a High-level Ecosystem Oversight Committee chaired by Justice Sangeet Lodha to audit the plants and draft a plan to restore the rivers. Based on the committee's scathing status reports, Justices Vikram Nath and Sandeep Mehta banned construction within 100 metres of the rivers' edges and hazardous industries within 500 m of the present path, or of the flood lines where available. The Court has also upbraided local authorities for failing to enforce pollution norms by citing the Court's own stay of a 2022 National Green Tribunal order imposing a compensation on the Rajasthan State Industrial Development and Investment Corporation and the local administration in Jodhpur and Barmer. The State regulatory apparatus has been dysfunctional: officials were aware of illegal mining operations in some areas, yet they regularised leases in those areas. Given the dereliction in play, the Court is correct to treat the possibility of collusion between State officials and industrial mafias seriously. Justice Mehta's observation that the State could be held liable under the 'polluter pays' principle is also a significant indictment of state failure, through negligence. Now, importantly, the State still has to restore the Jojari and attend to its wellbeing, which is relevant to a recent tendency by the Court to substitute itself for regulators rather than disciplining them into discharging their functions. This needs to change.

New Delhi's 'balancing act' risks eroding Russia's trust

The Shanghai Cooperation Organisation (SCO) summit in Bishkek, Kyrgyzstan, on August 31-September 1, 2026, and the BRICS summit in New Delhi on September 12-13, 2026, were remarkable. Iran received considerably more attention and support from members of both the groups, far more than anticipated. More consequential, however, was the shift in India's stance on the United States-Israel war against Iran and Russia's war in Ukraine.

Traditionally, the SCO and BRICS summits have offered the three major powers – Russia, India and China – an opportunity to demonstrate their mutual solidarity. For Russia, it is a platform from which it can signal to the West that it is not isolated and still has powerful allies. The expansion of the SCO and BRICS under China's leadership showcases its growing power to the West. For India, the SCO and BRICS have been useful for demonstrating its strategic autonomy. India has sought to remain welcome in both western and non-western forums, from the G-7 to the SCO and BRICS.



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A surprise pivot

At Bishkek, however, India sprang two surprises. First, it endorsed the Bishkek Declaration, which condemned the attacks on Iran as violations of international law, reaffirmed Iran's sovereignty and territorial integrity, and mourned the killing of its Supreme Leader, Ayatollah Ali Khamenei, in U.S.-Israeli air strikes on February 28, 2026.

Although the declaration did not name the U.S. or Israel, its meaning was unmistakable. India reaffirmed its opposition to military strikes on Iran by endorsing the Bishkek Declaration, which condemned the attacks and reiterated support for Iran's sovereignty and territorial integrity. Second, Prime Minister Narendra Modi made an unusually blunt appeal to Russian President Vladimir Putin to bring Russia's seemingly endless war in Ukraine to an end.

India has generally sought to avoid directly condemning the U.S. or Israel for their military actions against Iran, even though Iran, like India, is a member of both the SCO and BRICS. India had previously avoided explicitly accusing Israel of genocide in Gaza, although it had supported UN resolutions calling for a ceasefire and reaffirmed its commitment to Palestinian statehood. Indeed, New Delhi had emphatically declared that it stands with Israel, while often working in multilateral forums to soften language that is critical of Washington and Tel Aviv. But in both the Bishkek and Delhi Declarations, India signed on to very strong criticism of Israel as well as very strong affirmation of Palestinian rights and the two-state solution.

India's performative diplomacy could undermine its longstanding friendship with Moscow

One interpretation of this pivot in policy is that India may have signed on to the declaration to avoid alienating Mr. Putin and Chinese President Xi Jinping, and thereby jeopardising their attendance at India's BRICS summit. Without the two leaders, India's presidency of BRICS this year would have been a complete failure. Since Operation Sindoor (2025), India has faced a growing perception of diplomatic marginalisation. Meanwhile, Pakistan has gained diplomatic prominence through Washington's selection of Islamabad as a mediator in the conflict with Iran, while several Gulf nations have sought defence agreements with Pakistan. Consequently, India needed a successful and high-profile BRICS summit, which it got.

Contradictory shifts

While there is a shift in India's policy, New Delhi has not abandoned Washington entirely. Mr. Modi's public appeal to Mr. Putin to end the endless war in Ukraine can therefore be read as a counter-signal to Washington. Had the message been intended only for Moscow, it could have been conveyed privately. Delivering it publicly suggested that Mr. Modi was also reassuring the West. India may have joined the SCO and BRICS consensus on Iran to secure a successful BRICS summit, but it was now leaning more toward the western position on Ukraine than before.

These apparently contradictory shifts reveal the strain on India's foreign policy. New Delhi wants to support Israel while condemning attacks on Iran; buy Russian oil while aligning more closely with the West on Ukraine; and remain central to BRICS and the SCO while assuring Washington that it is not part of an anti-American bloc. Multi-alignment is becoming an increasingly difficult exercise in this era of change and turbulence. India's balancing act is unlikely to sit comfortably with either Moscow or Beijing.

Although Mr. Modi and Mr. Xi did not meet bilaterally at the SCO summit in Bishkek, they held talks on the sidelines of the BRICS summit in New Delhi. The meeting preceded Mr. Xi's high-profile G2 summit with U.S. President Donald Trump in Washington, which was held in September 2026. Neither summit declaration mentioned the war in Ukraine, avoiding criticism of Moscow even as Mr. Modi publicly urged Mr. Putin to end the conflict. India thus endorsed two declarations accommodating Russia's position without fully embracing it.

From Mr. Putin's perspective, this was the second time that Mr. Modi had used what might be called *'jumla'* (performative) diplomacy at Russia's expense. At Samarkand on September 16, 2022, Mr. Modi told Mr. Putin that "today's era is not an era of war". External Affairs Minister S. Jaishankar subsequently repeated that phrase in western capitals to demonstrate that India's purchase of discounted Russian oil did not signify political support for Russia's invasion.

The selectivity of these admonitions is difficult to ignore. Mr. Modi has not used comparable language toward Mr. Netanyahu, even as Israel has attacked several countries and territories across West Asia, many on multiple occasions, and has committed genocide in Gaza. Nor has Mr. Modi publicly directed the same admonition at Mr. Trump following American attacks on Iran. India itself attacked Pakistan during Operation Sindoor. Mr. Putin may therefore conclude that Mr. Modi's declaration that this is "not an era of war" reflects neither a consistent principle nor a universal opposition to war, but diplomatic messaging designed to reassure the West at Russia's expense.

Mr. Modi's latest appeal – that "endless war" must give way to an "end of war" – may reinforce that perception. Would Mr. Modi deliver the same message to Mr. Trump about the increasingly prolonged war with Iran – or to Mr. Netanyahu about his unending campaign of revenge in Gaza? How is Moscow interpreting this double standard?

Where ties with Russia stand

Russia, nevertheless, still needs India, at least for now, as a major purchaser of its oil and military equipment. At the same time, Moscow can see that New Delhi has steadily reduced its dependence on Russian arms by turning increasingly to France, Israel, and the U.S. The relationship remains important to both countries, but it is becoming increasingly transactional – and vulnerable to their diverging strategic priorities.

Russia is now leaning ever more toward its "trusted partner" – China, while Mr. Modi's performative diplomacy designed to stay in the good books of Washington risks undermining the long enduring Russia-India friendship.

India's next investment cycle will shape its carbon future

The United Nations Environment Programme's (UNEP) recent warning that the world could cross the 1.5°C warming threshold within the next few years is likely to renew calls for faster decarbonisation. India, too, will face this pressure, but its choices are still open. Much of the housing, industrial infrastructure, power networks and urban development needed for its next phase of growth is yet to be built.

NITI Aayog estimates that 86% of the building stock that India will have in 2070 is yet to be built. Total floor space will more than double, from about 18 billion square metres in 2020 to 42 billion by 2070. India's crude-steel capacity, at about 200 million tonnes in 2024-25, is targeted to reach 300 million tonnes by 2030-31. Cement output is projected to rise from about 427 million tonnes in 2024 to 660 million tonnes by 2030.



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The advantage of choice

The usual reading of these numbers is that India's infrastructure deficit makes decarbonisation harder. Demand for steel, cement and electricity will continue to rise even as India tries to reduce emissions. True, but only partly. The other side of the development deficit is that much of India's future capital stock remains a choice rather than an inheritance. Its future emissions will depend on what it chooses to build now and the technologies and standards that are embedded in that investment.

Mature economies face the opposite problem. The International Energy Agency (IEA) estimates that around 80% of the building stock that advanced economies will use in 2050 already exists. In the European Union, 85% of buildings predate 2000, around three-quarters perform poorly in terms of energy efficiency, and only about 1% are renovated each year. India has legacy problems too, but has more room to avoid creating them. It can build in efficiency from the outset and reduce the risks of stranded assets, premature retirements, or expensive retrofits later. In buildings, India need not wait for technology to catch up. The know-how is there, and so is the regulatory framework. The gap is in implementation.

Its future emissions will depend on what it chooses to build now and the technologies and standards incorporated into those investments

The Bureau of Energy Efficiency (BEE)'s State Energy Efficiency Index reports that the Energy Conservation Building Code (ECBC) 2017 has been notified by 27 States, but legally incorporated in only 21. BEE has since issued the Energy Conservation and Sustainable Building Code (ECSBC) 2024 for commercial and office buildings and the Eco-Niwas Samhita (ENS) 2024 for residential buildings. States should embed these standards in building permits, occupancy certificates, and grid connections, where non-compliance can carry a cost.

The issue in the power sector

Power presents a different problem. India has added renewable generation quickly, but the network to move it has not kept pace. The IEA estimates that transmission bottlenecks have held back around 60 GW of renewable capacity. The Ministry of Power's plan envisages roughly 590 GW of generation additions between 2023 and 2032. So, an unresolved transmission gap will become more serious as generation expands. As transport, cooling and parts of industry electrify, the gains from cleaner generation will depend on whether that power can be reliably moved to where it is needed. Transmission and storage, therefore, must be planned alongside, and in some cases ahead of, new generation and demand.

In steel and cement, the low-carbon transition is harder. The Ministry of Steel acknowledges that hydrogen-based green steel is not yet viable in India, while NITI Aayog places large-scale carbon capture in cement further down the transition path. Mandating technologies could raise costs and, worse, lock firms into processes that later prove uneconomic.

Public procurement could be a better instrument. The Union and State governments, Indian Railways, the National Highways Authority of India, and other public agencies buy large quantities of steel and cement and can help create an early market for lower-emission materials.

India already has a green-steel taxonomy, and the Ministry of Steel has proposed aggregated procurement through competitive bidding. Tenders could set emissions-intensity thresholds without prescribing the production route. Competitive bidding would then reveal the green premium while rewarding producers who find the most efficient ways to meet the thresholds. As the premium narrows with economies of scale and technological advances, the thresholds could be tightened. They could also be aligned with the emissions-intensity targets under the Carbon Credit Trading Scheme (CCTS), ensuring consistent policy signals.

Costs and risks

Finance can also skew these choices. The IEA estimates that India's cost of capital for grid-scale renewables is about 80% higher than in advanced economies. Since clean infrastructure is front-loaded in cost, that gap can make an otherwise viable project uneconomic. Targeted instruments such as payment-security mechanisms, partial credit guarantees, or concessional debt can ease financing or off-taker risk. But where the technology is still unproven, cheaper public finance will simply shift more risk onto the public balance sheet.

India cannot avoid the investment required to build the economy it still needs. What matters is what kind of capital the investment leaves behind. Where technology is mature, there is little reason to build inefficiency into assets that will last for decades. Where it is still evolving, preserving flexibility has value. Getting those choices right now can spare the country costlier corrections later.

The views expressed are personal

The SIR and its error margin



Nishtha Gupta

Technology policy researcher based in New Delhi

For nearly a month, Indians across the country have courted the ire and might of the state over a question no citizen in a democratic nation should have to ask. Their anger stems from the Special Intensive Revision (SIR) exercise which has removed more than 13 crore names from the rolls since June 2025. Over a quarter were claimed dead or duplicated, while most were marked as absent or shifted. At one booth in Delhi's Bela Estate, officials cut 728 out of 729 voters, citing address proof discrepancies. Who decided that so many names should come off the rolls and on what basis?

While the political answer is being debated on the streets and in Parliament, it is clear that the SIR has changed how a voter's place on the roll is decided. In 1995, the Supreme Court held that a name already on the roll should be presumed to be there legally. The SIR replaced that presumption with a data-matching exercise. The Election Commission of India (ECI) has relied heavily on software for a public task where a single error costs a citizen their vote.

Rolls from the last intensive revision (mostly in 2002) were hastily digitised from Bengali, Kannada, and other regional languages, auto-transliterated by the machine into English, and matched against new enumeration forms. Then a second layer of "logical discrepancies," such as a parent-child age gap of less than 15 years, was applied. Among those who failed the test were twins (age gap less than nine months), voters of the Oraon tribal group whose surname Xalxo was auto-translated to Khalkho, and those whose father's name was 'Md.' in one list and 'Mohammed' in the other. To reconcile the record, little logic or reasoning was applied. According to the Internet Freedom Foundation, these rules flagged 17% of West Bengal's electors and 41% of Madhya Pradesh's. A system that suspects four in 10 voters in a State needs to come under scrutiny.

Such sweeping pattern-matching guarantees false positives. In West Bengal, 91% of exclusion appeals decided by the tribunals were reversed; a reversal rate that high should worry whoever designed the process. And by the time their eligibility was restored, these people had already lost the opportunity to vote in the 2026 Assembly election.

Drawing the line

This is why it matters where the line is drawn for who is "suspicious". How many eligible voters are risked to remove one ineligible voter? Legally, Election Registration Officers (EROs) determine where the limit lies. Yet in the SIR, the ECI has never revealed what the acceptable margin of error is or the data itself. Two of the three Election Commissioners are on record objecting to the centralisation of the database and the restrictions imposed on local officers' authority. In Goa, officers heard 97 flagged voters and found all eligible, but couldn't restore them because the software had no option to reverse the earlier action.

The ECI claims that automation is only meant to assist human decisions. This would be true if officers actually had the chance to evaluate every flagged entry. In Delhi, each booth-level officer received about 200 notices to verify within a very tight deadline. When the queue becomes practically unmanageable, the machine's verdict on an individual case inevitably becomes the State's verdict.

The citizen then pays to undo this through hearings and lost wages, or is forced to remain disenfranchised if unable to appeal. In West Bengal, the damage went beyond voting when a June State order marked the ration cards of SIR-removed voters as inactive. The SC has since restored access to food. India has sold its digital public infrastructure to the rest of the world on the premise of inclusion at scale. Built on faulty data, the same architecture excludes at scale, and an error in one register follows a person into the next.

This doesn't mean that the rolls shouldn't be cleaned. What must change is where the risk tolerance lies. A dead voter left on the roll is a clerical error. A living voter who is struck off loses their constitutional right. The ECI should disclose the matching rules and error rates and open itself to an independent audit. No name should be struck off until an officer with actual time and authority verifies the real person behind the data.

The model is already travelling. In January, Chief Election Commissioner Gyanesh Kumar offered the ECINet platform to help other countries build their own versions. In August, U.S. President Donald Trump cited India's photo ID rules to push for stricter U.S. voter laws. Whatever standard the world's largest democracy accepts for electoral error, other nations will follow.

A dead voter left on the roll is a clerical error. A living voter who is struck off loses their constitutional right

Are coronary stents necessary?



Dr. Enas A. Enas

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Dr. Basil Varkey

Professor Emeritus of medicine

Keralam Chief Minister V. D. Satheesan's recent remarks questioning "unnecessary" angioplasty procedures have drawn objections from medical organisations concerned that it could undermine public confidence in doctors. But the Chief Minister's remarks demand more than a reflex response; it requires a thoughtful, honest response built on up-to-date evidence based on clinical and research studies.

Percutaneous Coronary Intervention (PCI), by balloon angioplasty and stent placement, has transformed cardiovascular care. During an acute heart attack caused by a suddenly blocked coronary artery, prompt angioplasty and stents can restore blood flow, save heart muscle and improve survival. There is no debate about this approach and second opinions are unnecessary as time is of essence in such an emergency. Stents also offer substantial relief to selected patients with persistent activity-limiting angina (chest pain caused by lack of oxygen-rich blood).

There is a crucial distinction between stenting as a life-saving measure in the above situations to medical treatment via stenting because a narrowing of blood vessels is seen on an angiogram. A coronary angiogram shows the anatomy of the arteries; it does not, by itself, establish that a narrowing requires intervention. The decision to intervene and how to intervene must take account of the patient's symptoms, the severity of the blockage, its effect on blood flow, the overall pattern of coronary disease and the expected benefit of intervention.

The landmark ISCHEMIA trial, published in *The New England Journal of Medicine* in 2020, enrolled 5,179 patients at 320 sites in 37 countries with stable coronary artery disease and moderate to severe ischemia (restricted blood flow to the heart). The patients were assigned either to an initial invasive strategy – angiography followed by angioplasty or coronary bypass surgery – or to a conservative strategy of optimal medical treatment, with revascularisation reserved for those who had an acute event or angina. After a median follow-up of 3.2 years, there was no significant difference between the two groups in composite outcomes (cardiac arrests, hospitalisation for unstable angina or heart failure) and all-cause and cardiovascular mortality were similar. After a follow-up of 5 years, there was still no significant difference between the two groups in composite outcome or mortality. The core lesson is that most patients with stable coronary artery disease with ischemia on optimal medical treatment would do as well as those who undergo stenting or bypass surgery.

Concerning scale

The growing scale of PCI in India deserves careful examination. National Interventional Council figures show that annual PCI procedures rose from approximately 1,13,000 in 2008 to more than 7,26,000 in 2024 – an increase of more than six-fold. In contrast, PCI procedures in the U.S. fell by about 40% between 2003 and 2016, reflecting progress in prevention and treatment.

These figures do not by themselves prove that stents are overused in India considering its large population, the substantial burden of premature coronary disease, the socio-economic progress in the past decades and the improved access to cardiac care centres. Nevertheless, the divergence in trends between the two countries raises an important question: how much of the increase in PCI is a reflection of fulfilling a genuine clinical need, and how much of it is a reflection of differences in practice patterns, financial incentives and gaps in knowledge on the efficacy of optimal medical treatment?

Optimal medical therapy, based on evidence-based guidelines, is the foundation of care for all coronary artery disease patients. The elements of optimal therapy include LDL-cholesterol lowering, appropriate anti-platelet treatment, blood-pressure control, evidence-based diabetes management and lifestyle changes. A coronary stent treats a specific narrowing; it does not cure the underlying disease of atherosclerosis or eliminate the risk of future heart attacks. Patients remain vulnerable to disease elsewhere in the coronary arteries and must continue preventive treatment.

The way forward is neither condemnation of cardiologists nor unquestioning acceptance of every recommendation for angioplasty. Most doctors strive to act in their patients' best interests, but no profession is immune to inappropriate practice. The ultimate test of healthcare is not how many procedures are performed, but how many lives are saved and whether patients receive the right treatment at the right time at an affordable cost.

There is a crucial distinction between stenting as a life-saving measure to medical treatment via stenting because a narrowing of blood vessels is seen on an angiogram

Text&Context

THE HINDU

What was leaked in the Censor Board data breach?

What impact could the leak have? When was the leaked data taken down? What is the status of the probe?

Aroon Deep

The story so far:

A data breach at the Central Board of Film Certification (CBFC) left hundreds of applications' data exposed online, *The Hindu* found. The data comes from over 500 titles, of which dozens are feature films. The leak comes from the censor board's e-Cinepramaan portal, used by filmmakers to submit their applications. The records ranged from December 2025 to August 2026.

What were the leaked documents?

The leaked documents included shooting scripts, with detailed shot-by-shot translations and other annotations. These scripts appear to have been specifically prepared for the censor board's review of these films – one script even included labels on swear words used over the course of the film and other clarifications on dialogue.

The breached data also included a summary of cuts that were ordered by the CBFC. This is supposed to be public information anyway under Section 5A(2) of the Cinematograph Act, 1952, but the censor board put this data behind an "under maintenance" message on its website.

In what could be the most sensitive of the exposed documents, film distribution agreements – both from international filmmakers distributing into India and Indian filmmakers authorising dubbed versions in a different part of the country – were exposed. At least three such agreements had unredacted or improperly redacted information on what fees were paid to license a film for distribution in India. *The Hindu* is withholding specifics here.

"Commercial agreements in the film industry often contain carefully negotiated provisions governing creative control, casting, remuneration, profit-sharing, intellectual property rights, promotional obligations, and even specific conditions concerning a film's storyline, portrayal or production," Vasundhara S., managing partner at Verum Legal told *The Hindu*.

"When commercially sensitive or specially negotiated creative terms enter the public domain, the confidentiality underpinning these agreements is compromised," Ms. Vasundhara said. "Such disclosures can weaken negotiating positions, reveal commercially valuable information, influence future contractual negotiations and potentially expose parties to disputes arising from confidentiality obligations."

"If I am a competing party or a counterparty, I'd be in a much better position to negotiate now that I have confirmation as to what terms these guys would agree to for certain types of works, beyond what is generally known in the market," Abhinav Hansaraman, a commercial lawyer in Delhi said.

Where was the data posted?

The documents were posted as dockets that contained practically everything but a copy of the film itself, and posted on the site of the non-profit Internet Archive portal. The data dump, most of it was posted on a single day, lay on the site for days, open to anyone to review and download. It is unclear if the data dump's existence was known to anyone.

The majority of the documents were posted earlier this month, but were all taken down on October 12. The user, whose display ID was former CBFC chairman Prasoon Joshi's name in Russian, also deleted their account. *The Hindu* was unable to contact them.

How did CBFC respond to the breach?

Last week, *The Hindu* contacted the CBFC for comment on the breach. The censor board did not appear to have immediate information on the breach. A week after being provided granular details of what documents were leaked from which film application, CBFC chairperson Shashi Shekhar Vempati said in a statement that it was looking into the breach. "CBFC has reviewed the information shared regarding alleged unauthorised access to CBFC's e-Cinepramaan Portal," the CBFC said. "However, based on the information examined so far, no trace has been found of the certification-related documents that were claimed to have been accessed."

The Internet Archive told *The Hindu* in a statement that the account in question was taken down by the user themselves. "As a matter of caution, a complaint has been filed with Cybercrime, and CERT-In has also been notified of the allegations and provided with the information shared," the CBFC added in its statement to *The Hindu*. CERT-In is India's computer emergency response team, which is tasked with handling data breach and hacking incidents.

"The technical team has also conducted a thorough analysis of the servers and databases to assess whether any external attacks may have compromised their security," CBFC added. "The findings of this analysis are being considered alongside the ongoing forensic examination."

On October 4, the e-Cinepramaan portal was unavailable for hours. The user stopped uploading any further documents shortly thereafter. No individual filmmaker's account seems to have been compromised, the CBFC said in its statement, and committed to a "thorough security audit" of the e-Cinepramaan portal.

Science successful only when society benefits from it: Modi

PM stresses on importance of taking science from laboratory to land; he says India is no longer playing catch up and emerged as leader in scientific, technological developments in 21st century

Jacob Koshy
NEW DELHI

Prime Minister Narendra Modi on Tuesday said science succeeds only when it travels from the laboratory to the land and improves people's lives.

Addressing the Emerging Science, Technology and Innovation Conclave (ESTIC), he said it must not become "merely a forum for discussion" but should deliver measurable outcomes.

Mr. Modi said India's science and technology story has long been one of catching up with other countries, but in the 21st century the country has emerged as a leader in scientific and technological developments. He said, "The discovery of zero showed that India has a culture of discovery. Chandrayaan's landing near the lunar south pole, Mangalyaan's success on the first attempt, and the vaccines developed during the COVID-19 pandemic are examples of this."

The audience at the conclave comprised three Nobel Laureates, top technocrats of India's scientific establishment, startup founders and scientists from both India and abroad.


Prime Minister Narendra Modi at the Emerging Science, Technology and Innovation Conclave 2026 in New Delhi on Tuesday. PTI

Focus on research

Mr. Modi said a country's technological capabilities depend on its research. He said efforts to "speed up" the Anusandhan National Research Foundation (ANRF) work are underway. Referring to research funding, he cited a commitment of ₹1 lakh crore, including ₹12,000 crore for taking research from the laboratory to the market, with private investors partnering equally. He also referred to new commitments of ₹24,000 crore.

The number of patents filed in India had risen from about 50,000 to 1.5 lakh, Mr. Modi said, describing this as evidence of a strengthening culture of innovation rather than merely an increase in number.

Science in the age of AI

On artificial intelligence (AI), he posed what he called an "important question" about how to exercise scientific judgement in the age of AI.

"Science works by questioning paradigms and looking at problems with fresh eyes," he said. "AI can perform routine tasks quickly and raise productivity, but should not limit scientists' thinking or independent thought, and its answers must meet the accuracy standards of scientists."

Referring to the birth anniversary of nuclear scientist Homi J. Bhabha on October 30, Mr. Modi said a developed India will need large quantities of nuclear energy. He noted that the 500 MW Prototype Fast Breeder Reactor (PFBR) has reached first criticality, strengthening the country's long-term energy position.

He also pointed to Indian space start-ups, a rocket made in India that placed a satellite in orbit on its first attempt, and the country's satellite docking capability. Indian-made AI chips are reaching other countries, an AI model has been built domestically, and work is underway on battery technology based on Indian raw materials, he said. The needs of sectors ranging from nuclear energy to space to AI and semiconductors differ, he said, but India's emerging technologies are rising together.

Describing India as a young country with vision and imagination, Mr. Modi said it needs a generation capable of leading in technology. He summed up the guiding principle as: "We will do it."

Nobel winner backs GM crop innovation in Indian labs

Jacob Koshy
NEW DELHI

Richard J. Roberts, the 1993 Nobel laureate in Physiology or Medicine, on Tuesday dismissed as "nonsense" the idea that Indians adopting genetically modified crops would become dependent on foreign companies.

The work can be done in, and owned by, Indian laboratories, he said.

Dr. Roberts was delivering a plenary address at the second Emerging Science, Technology and Innovation Conclave (ESTIC) at Bharat Mandapam.

In his address, he rejected the term 'genetically modified' as "very misleading" and referred to such crops throughout as "biotechnologically modified," or BM, arguing that "every life form on this planet has been genetically modified by evolution."

Asked about fears that an American corporation such as Monsanto (whose operations were subsumed into that of agrochemicals giant Bayer) could control the seed supply and erode the country's independence, Dr. Roberts replied: "I think it's nonsense. You can do improved agriculture here in the labs in India and own everything here in India. You don't have to go to America or Europe or anywhere else to make CRISPR-modified plants. Anybody can do it. You go to a lab, learn how to do it – maybe it takes you six to eight weeks – and then you can do it. It's really very straightforward


Richard J. Roberts

because we've come a long way in knowing how to improve crops. But you can own it here in India. "

CRISPR is a gene-editing tool in which a guide RNA steers the Cas9 enzyme to a chosen spot in an organism's DNA, where it cuts, letting scientists alter or insert genes at that site. The foundational CRISPR-Cas9 patents, however, are embroiled in an intellectual property dispute.

A divisive saga

GM food crops have a contested history in India. In 2010, then Environment Minister Jairam Ramesh imposed an indefinite moratorium on Bt brinjal, a genetically modified variant of the vegetable. In July 2024, the Supreme Court delivered a split verdict on GM mustard's environmental release, but unanimously asked the Centre to frame a national policy on GM crops. Golden Rice, engineered in 1999 by Ingo Potrykus and Peter Beyer to carry a vitamin A precursor in its grain, has also proved divisive.

Indian researchers reported in the peer-reviewed PLOS One journal in 2017 that backcrossing one Golden Rice transgene event into an Indian variety had yielded stunted, low-yielding plants, unfit for cultivation despite their high provitamin A content.

Asked whether Golden Rice could disturb India's agricultural systems, citing the invasive spread of eucalyptus, Dr. Roberts said he was not an expert in agriculture, underlining that there are people out there who truly are experts who can answer the questions much better than he can.

India permits Bt cotton for commercial cultivation, but no GM food crop.Kavitha Kuruganti, who leads the Coalition for a GM-Free India, has previously argued that the yield claims for GM mustard, which was cleared by a technical body but whose release subsequently stalled, are not supported by the developers' own data. India needs a biosafety policy rather than a policy on GM crops, she said.

Dr. Roberts framed GM as a question of food and nutritional security. "Food is medicine," he said, for people who cannot afford supplements or drugs. Crops eaten in developing countries, he argued, never received the breeding investment Western staples did because their farmers could not pay for it. "Golden Rice is not priced above ordinary rice, and several Indian government institutes hold licences for it," he said, and called Bt cotton a "success in India."

Delhi invites Pakistan Minister for SCO Civilisational Dialogue debut

Suhasini Haidar
Sreeparna Chakrabarty
NEW DELHI

India will hold the first "Civilisational Dialogue Forum" of the 10-nation Shanghai Cooperation Organisation (SCO) grouping, that includes Pakistan, from November 20-22 in Delhi, sources confirmed to *The Hindu*.

The dialogue, which will be organised by the Ministry of Culture and the Ministry of External Affairs, was proposed by Prime Minister Narendra Modi at the SCO leaders' Summit in Tianjin last year and adopted by the SCO Council.

According to the sources, the government has invited Pakistan's Minister for National Heritage and Culture Aurangzeb Khan Khichi, for the event. Culture Ministers from the 10 member states, including China, Russia, Iran, Belarus and Central Asian states, 15 Dialogue partners and observer countries


Aurangzeb Khan Khichi

have been invited.

The event, which will be inaugurated by Mr. Modi, will be held at Delhi's iconic Purana Quila (Old Fort), a 16th Century fort built by Emperor Humayun. The fort is also the site of many excavations showcasing India's ancient past spanning nearly 3,000 years, from the Pre-Mauryan, Mauryan, Shunga, Kushana, Gupta, Post-Gupta period to medieval-era Rajput, Sultanate and Mughal periods, officials said, explaining its importance.

According to diplomatic sources, Pakistan has not so far confirmed who will lead its delegation to the dialogue forum. If Mr. Khichi attends, this will be the first time India will host a Minister from Pakistan since the 2023 SCO Foreign Ministers meeting, when the then Pakistan FM Bilawal Bhutto had travelled to Goa, and the first since the four-day conflict between India and Pakistan in May 2025.

While India has refused to take part in meetings of South Asia Association for Regional Cooperation (SAARC) in Islamabad due to differences with Pakistan since 2016, it has joined SCO meetings there in the past, including SCO-Regional Anti-Terror Structure (RATS). In September, India sent SCO National Coordinator Alok Dimri to Islamabad to attend the first coordination meeting for Pakistan's SCO Presidency and events leading up to the leader's Summit in Pakistan, scheduled for August 2027. In November 2024, External Affairs Minister S. Jaishankar had represented India at the SCO Heads of Government (HoG) Meeting, and all eyes will be on who will represent India at the SCO Heads of State Summit next year.

"India deals with SCO as an entity, as an important organisation," MEA Secretary (West) Sibi George said in response to a question from *The Hindu* at a briefing on August 27 about whether PM Modi would travel to Pakistan for the summit. "As far as our engagement with individual countries is concerned, a decision will be taken by the appropriate authority at the appropriate time and our appropriate level of participation will be decided accordingly," he added.

For the forum, each participating State has been invited to nominate a delegation comprising the Minister responsible for Culture and up to four government representatives.

Economic policy cannot outweigh citizens' lives: SC on medicine prices

Aaratrika Bhaumik
NEW DELHI

The Supreme Court on Tuesday said "economic policy" could not take precedence over the lives of citizens as it questioned the Centre over its approach to regulating the prices of essential medicines.

A Bench of Justices Vikram Nath and Sandeep Mehta questioned Additional Solicitor-General Aishwarya Bhati, appearing for the Centre, over the failure of the National Pharmaceutical Pricing Authority (NPPA) to cap trade margins on essential medicines despite concerns over excessive markups. It also sought to know why such a ceiling had been imposed on certain anti-cancer drugs only after the court's intervention. Earlier this month, the NPPA, the government's drug-price regulator, capped trade margins on 137 non-scheduled anti-cancer medicines at 30% of their maximum retail price (MRP). The revised prices are set to take effect from November 9.


SC questioned the Centre over its approach to regulating medicine prices. FILE PHOTO

"For so many years, why were you silent? Suddenly, you decided to cap trade margins of cancer drugs?... Don't force us to ask harsh questions," the Bench remarked.

The oral observations came during the hearing of petitions filed by advocate Kishan Chand Jain and paediatric surgeon Sanjay Kulshrestha seeking stricter enforcement of price controls under the Drugs (Prices Control) Order, 2013, (DPCO) to curb disproportionate margins across the pharmaceutical supply chain.

Ms. Bhati, however, told the Bench that "complete government control" over essential medicines could adversely affect the market. "We want investments to also come here... We can make these interventions from time to time, but complete control is not the answer," Ms. Bhati submitted.

The Bench also questioned why the 16% retailer margin prescribed under the DPCO should not apply to all essential medicines and sought the basis for classifying drugs as scheduled or non-scheduled. While scheduled formulations are subject to government-fixed price ceilings, non-scheduled medicines remain outside this framework.

"How do you decide which medicine is essential and which is not? Why should all medicines not be essential? That is where the game is... the NPPA seems to be totally defunct," Justice Mehta remarked.

Senior advocate Mukul Rohatgi, appearing for an association of hospitals, defended the steep markups on essential medicines, including anti-cancer drugs, contending that capping such margins would make hospital operations financially unviable. The Bench, however, dismissed the contention, remarking that patients could not be subjected to "dacoity". It reserved its order in the matter and gave the parties two weeks to file their submissions.

InvITs look at a bigger slice of India's vast household savings

Lalatendu Mishra
MUMBAI

India's infrastructure investment trusts (InvITs) are attracting more investors, creating a potential avenue to channel domestic savings into roads, power transmission networks and other income-generating infrastructure assets.

The number of investors in listed InvITs rose to 6.53 lakh in the first quarter of FY27, while assets under management (AUM) increased to ₹7.3 lakh crore, according to the Bharat InvITs Association (BIA).

Distributions to unitholders during the quarter stood at ₹5,923 crore, taking cumulative payouts since the sector's inception beyond ₹97,000 crore.



Lower entry barriers are an attraction to invest in InvITs.

The growth comes as India faces mounting financing requirements for infrastructure expansion. While government spending and foreign investment remain important, household savings and investments held by mutual funds, insurers and pension funds could provide a larger pool of long-term capital, according to industry executives.

InvITs offer a route to connect these savings with operational infrastructure assets. Investors buy units in trusts that own such assets and receive distributions from the income generated, without directly financing or managing the underlying infrastructure.

"India's next infrastructure financing challenge is creating efficient pathways for enormous pool of local savings to reach infrastructure," said Gaurav Chandna, ED and Joint CEO, Vertis Infrastructure Trust.

The model also helps developers monetise completed, revenue-generating assets, freeing up capital for new projects.

India needs to strengthen capabilities across value chain: Kamaljeet Singh

Director-General of the Semiconductor Laboratory highlights foreign dependence for critical inputs, and calls for stronger R&D, manufacturing infrastructure and specialised skills; he also cites SCL's efforts to make semiconductor design education more accessible through open-source tools

The Hindu Bureau
CHENNAI

India needs to strengthen capabilities across the semiconductor value chain, from manufacturing and materials to research and specialised skills, to reduce dependence on foreign technologies and build a resilient domestic ecosystem, Dr. Kamaljeet Singh, Director-General of the Semiconductor Laboratory (SCL), Mohali, said on Tuesday.

Speaking at the inaugural session of the Semiconductor Summit 2026, organised by SRM Institute of Science and Technology in association with *The Hindu*, he said India had a significant presence in semiconductor design but remained dependent on foreign suppliers for critical inputs, including electronic design automation (EDA) tools.



Michael Hasper, Consul-General of the Federal Republic of Germany; L.V. Navaneeth, CEO, The Hindu Group; Kamaljeet Singh, Director-General of the Semiconductor Laboratory (SCL), Mohali; and C. Muthamizhchelvan, Vice-Chancellor of SRM Institute of Science and Technology. S.R. RAGHUNATHAN

Mr. Singh said the country needed to overcome several challenges to become a technological power. "Technology is one of the challenges, talent is one of the challenges, resources is one of the challenges, time is the challenge, acceptability is the challenge and capital is the challenge," he said.

He said India needed to build capabilities not only in chip design and fabrication but also in the supporting supply chain, including manufacturing equipment, raw materials, chemicals, gases and packaging materials. Semiconductor supply chains were concentrated in a small number of countries, making access to these inputs critical, he said.

He highlighted the need to connect research with manufacturing, arguing that developing semiconductor technologies required researchers, pilot lines, manufacturing facilities and industry participation. "You need a lot of researchers to come into place, you need a manufacturing unit, you need a lot of technology to be developed to attain the system design and architecture," he said.

Mr. Singh said the India Semiconductor Mission had supported the development of domestic capabilities. Under the first phase, ₹76,000 crore had been pumped in, with 12 projects approved, five of which had been operationalised and another expected to become operational, he said.

The second phase of the mission had six pillars, including chip design, manufacturing, materials and machinery, with research and development and skill development added as two further areas of focus, he said. He stressed that India needed to strengthen specialised engineering and manufacturing capabilities alongside its existing design talent.

Mr. Singh highlighted SCL's efforts to make semiconductor design education more accessible through open-source tools. The laboratory had integrated its process design kit (PDK) with open-source tools so that students unable to afford commercial EDA software could learn chip design.

He said SCL was also working with academic institutions and startups and had introduced training programmes, internships and faculty development initiatives.

SCL also supplied chips for space programmes, including Chandrayaan-3, and was working towards developing integrated solutions combining sensors and chips with software and firmware.

SEMICONDUCTOR SUMMIT

Chip war exposes semiconductor vulnerabilities, industry leaders say

They stress the need for India to identify areas where it can develop technologies of its own while building domestic manufacturing capabilities and diversifying sources of supply, as countries increasingly seek to secure access to semiconductor technologies, materials and manufacturing capacity

The Hindu Bureau
CHENNAI

As geopolitical tensions reshape the global semiconductor industry, India needs to develop its own technology capabilities, reduce critical supply-chain dependencies, and build resilience through international partnerships, industry leaders said at a panel discussion on 'The chip war: technology, geopolitics and India's challenges' at the Semiconductor Summit 2026 here.

The panellists said India needed to identify areas where it could develop technologies of its own while building domestic manufacturing capabilities and diversifying sources of supply, as countries increasingly seek to secure access to semiconductor technologies, materials and manufacturing capacity.

Sharat Kaul, India Representative, The International Electronics Manufacturing Initiative (iNEMI), said the country should not dismiss outsourced semiconductor assembly and testing (OSAT) projects as an inadequate starting point. Semiconductor manufacturing requires specialised equipment, materials, talent and infrastructure, making it necessary to build the ecosystem progressively. "We have to crawl before we walk. We have to walk before we run," he said.



Sharat Kaul, India Representative, The International Electronics Manufacturing Initiative; Sandeep Garg, CEO and Board Member, Indichip Semiconductors Limited; Ravi Bhatkal, vice-president strategy, MacDermid Alpha Electronics Solutions; and Yogan Senthilkumar, chief technology officer, iVP Semiconductor at the Semiconductor Summit on Tuesday. S.R. RAGHUNATHAN

Mr. Kaul said semiconductor sovereignty did not mean producing everything domestically.

"Sovereignty doesn't necessarily mean that we do everything ourselves. It means that we build reliance and resilience. A bigger part is really resilience," he said, citing the importance of maintaining multiple supply-chain options through partnerships with friendly nations.

He also identified advanced packaging, which integrates multiple chips on a single substrate, as an opportunity for India to narrow its technological gap with established semiconductor manufacturing countries.

"The tipping point in the industry is going to be advanced packaging," he said, pointing to the potential of two-and-a-half-dimensional and three-dimensional technologies amid geopolitical tensions and efforts by countries such as the United States to bring advanced manufacturing within their borders.

Sandeep Garg, CEO and Board Member of Indichip Semiconductors, said India needed to manufacture semiconductor devices required by its own industries instead of relying on imports.

"The time has come for manufacturing and we have to manufacture our own devices," he said.

Drawing on his industry experience, Dr. Garg said major semiconductor companies had not established fabrication facilities in India despite being present in the country for decades. India needed to create greater domestic value addition rather than assume its large market alone would attract manufacturing investment, he argued.

Indichip is setting up an integrated device manufacturing facility covering fabrication, assembly and testing, and final modules, with an investment of more than ₹15,000 crore, he said. The company focuses on silicon carbide power devices used in sectors including energy, electric mobility, railways, defence and data centres.

Dr. Garg estimated that India's silicon carbide market could grow from about ₹13,000 crore in 2030 to ₹1 lakh crore by 2035.

However, domestic manufacturing faces gaps in essential inputs. A silicon carbide fabrication facility uses 37 gases, of which India produces only four, while it requires 42 chemicals, most of which are not manufactured domestically, he said.

Dr. Garg called for the semiconductor sector to be granted industry status and for the government to address the goods and services tax burden on supporting ecosystem players. With semiconductor manufacturing projects taking three to five years, he wondered how suppliers could absorb the tax burden during the long lead time.

Key concerns

Ravi Bhatkal, vice-president of strategy at MacDermid Alpha Electronics Solutions, said India needed to develop domestic capabilities in mining, refining and materials production while securing supplies from multiple countries for inputs it could not produce locally.

Yogan Senthilkumar, chief technology officer of iVP Semiconductor, said semiconductor education needed to extend beyond chip design to manufacturing, assembly, testing, materials and equipment.

The panellists also discussed whether India could compete with countries investing substantially more capital in semiconductors.

WHAT IS IT?

Glaciers ending in lakes contribute to a higher ice loss

Press Trust of India

Glaciers ending in lakes, oceans or ice shelves account for over 3% of the world's glaciers but lost 134 billion tonnes of ice annually between 2000 and 2019, contributing to more than half of all glacier ice loss, a new study has found.

Environmental scientists led by the UK's University of Birmingham and the University of Leeds found that glaciers ending in lakes are generally much larger than those ending on land, together covering more than half of the world's glacier area.

The findings, published in *Nature Communications*, also showed that lake- and ocean-ending glaciers flow on average 10% and 44% faster than those ending on land, respectively, and lake-ending glaciers lose around 57% more ice than land-ending glaciers, relative to their size.

"Glaciers ending in lakes really stand out -- they're losing ice particularly quickly. We don't yet know exactly why. It could be the lakes themselves, the size and shape of the glaciers, or a bit of both. Untangling that is the next big question, especially as glacier lakes are becoming more common in a warming climate," said author Joseph Mallalieu.

The researchers explained that lakes can affect glaciers in ways that may

Ice melts at the Skuta glacier near Kamnik, Slovenia. AP

contribute to a higher ice loss. Lake-ending glaciers can lose large chunks of ice from the glacier front as icebergs, a process known as calving, and lake water can melt the ice below the waterline, both helping to speed up ice loss, they said.

Models used for projecting glacier change do not yet fully account for glaciers ending in lakes, the researchers added.

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